Retrieval Augmented Generation (RAG) — Lab Knowledge Guide

What is RAG?

Retrieval Augmented Generation (RAG) is an architecture in which an AI application retrieves relevant information from an external knowledge source and provides that information to a language model as context for generating a response.

RAG is useful when the required information is private, domain-specific, frequently changing, or too large to place directly into a prompt.

RAG Workflow

- User asks a question.
- The application converts the question into a search request.
- A search service retrieves relevant document content.
- Relevant content is supplied to the language model as context.
- The model generates a response based on the retrieved context.
- The application can display citations or source information when supported.

Documents and Chunking

Large documents are commonly divided into smaller chunks before indexing. Chunking helps the search system retrieve focused passages rather than entire documents.

Good chunking should preserve enough context to understand the passage while keeping individual chunks focused. The best configuration depends on document structure and the application.

Azure AI Search in RAG

Azure AI Search can provide indexing and retrieval capabilities for RAG applications. It can store searchable representations of document content and return relevant results for a user's query.

A common lab flow is: upload documents to Azure Storage → create a data source → create an indexer or indexing pipeline → create a search index → configure semantic or vector retrieval where appropriate → connect the search capability to an AI application.

Knowledge Sources

A knowledge source identifies information that an AI application can use when answering questions. In a RAG solution, the knowledge source can be backed by indexed documents and a search service.

The quality of the final answer depends on both retrieval quality and model behavior. If relevant information is not retrieved, the model may not have enough context to answer correctly.

RAG Testing

Test with three types of questions: questions clearly answered by the documents, questions requiring information from multiple documents, and questions that are outside the knowledge base.

A grounded agent should distinguish between information found in its knowledge and information that is unavailable. A useful instruction is: If the answer cannot be supported by the provided knowledge, say that the information was not found rather than inventing an answer.

Example

Suppose the knowledge base contains an Azure AI Foundry guide and an Azure services guide. A user asks: What Azure service can be used to retrieve information for a RAG application?

The retrieval system may return passages describing Azure AI Search. The language model then uses those passages to produce a concise answer, such as: Azure AI Search can be used to index and retrieve relevant content for a RAG application.